A nilpotent cellular automaton that remembers under noise

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Abstract

Does every positive zero-range perturbation of a nilpotent cellular automaton have a unique attracting stationary law? We construct a counterexample on $\mathbb{Z}^3$, answering a question of Marcovici, Sablik and Taati. Each site carries a bit and a bounded countdown. The deterministic rule reaches the all-zero configuration after a fixed number of steps, while independent uniform replacements renew the countdowns and allow a horizontal bit plane to retain its initial value. With stationary countdowns, the plane is exactly a north-east-centre majority process with independent gates and independent replacement faults. We calculate the gate law and apply Toom's stability theorem in the quantitative version of Gács to obtain two translation-invariant stationary laws. For the same fixed automaton, a dependency-path estimate proves uniform exponential ergodicity at sufficiently small and sufficiently large noise rates. We also construct, for every prescribed separation, a nilpotent automaton, a set of replacement events at pairwise distance exceeding that separation, and two initial configurations whose bits differ on a whole plane at all times under those common events.

The countdown also lifts other rules. Applied to the computable family of positive-rate automata on $\mathbb{Z}^8$ whose ergodicity at a fixed noise rate is undecidable (Douglas and Mghirbi, *Forgetting under noise is undecidable*), it gives a computable family of radius-one cellular automata on $\mathbb{Z}^9$, all nilpotent within the same number of steps, with alphabets that grow with the machine. At every rate in $[\varepsilon_0/2, \varepsilon_0]$, where $\varepsilon_0 = 490000^{-99}$, uniform replacement noise makes the member built from a Turing machine M ergodic if and only if M halts. For every n, we also construct one nilpotent automaton on $\mathbb{Z}^3$ with n disjoint intervals of noise rates that carry two translation-invariant stationary laws, alternating with intervals of uniform exponential ergodicity. Whether a single nilpotent rule can have infinitely many such windows is open.

Erasure in two steps already suffices, while erasure in one step forces ergodicity under every zero-range noise. We give a four-symbol cellular automaton F on $\mathbb{Z}^3$ with $F \circ F$ constant whose uniform replacement perturbation has two translation-invariant stationary laws at every rate in $[2^{-44}, 2^{-42}]$ and is uniformly exponentially ergodic at sufficiently small and sufficiently large rates. Each site reads a seed, the word of 2^{51} auxiliary symbols above it, which every deterministic update resets to zero. Exact rational seed partitions give a compiler: for fixed rationals $0 < \varepsilon < 2\rho$, it maps every attractive binary probabilistic cellular automaton on $\mathbb{Z}^d$ with rational transition probabilities in $[\rho, 1-\rho]$ computably to a cellular automaton on $\mathbb{Z}^{d+1}$, nilpotent in two steps, whose uniform replacement perturbation at rate ε is ergodic, uniformly exponentially ergodic or has two translation-invariant stationary laws exactly when the original automaton has that property.

AI systems assisted with this work; see "How this work was done."

1 Introduction

A nilpotent cellular automaton erases every initial configuration in a bounded time. It is natural to ask whether adding independent noise can only reinforce this loss of memory. Marcovici, Sablik and Taati asked whether every perturbation of a nilpotent cellular automaton by positive zero-range noise is ergodic [2, Problem 6.5]. We give a negative answer using a bounded countdown and Toom's majority rule. Our example is three-dimensional, and its alphabet has $2(H+1)$ symbols; the independence of the gates uses the third coordinate. We do not address dimensions one and two.

The distinction between a fixed small noise rate and a noise rate tending to zero is essential. Every fixed nilpotent rule remains ergodic under sufficiently small perturbations [2, Theorem 3.9]. In our example, noise must arrive often enough to renew a countdown before it expires. A long countdown makes this renewal effective at a noise rate still small enough for majority correction to preserve a bit. At still smaller rates the countdowns expire and erase the bits.

Here is the resulting behavior for one fixed rule.

Theorem A. There is a finite-alphabet, radius-one cellular automaton on $\mathbb{Z}^3$ that is nilpotent and whose independent uniform replacement perturbation has at least two translation-invariant stationary laws at a positive noise rate. The same rule is uniformly exponentially ergodic at all sufficiently small and all sufficiently large positive noise rates.

The explicit rule is given in Section 2; Theorems 4.2 and 5.1 prove the assertions. A purely deterministic statement isolates the renewal mechanism.

Theorem B. For every $d \geq 0$, there is a nilpotent cellular automaton on $\mathbb{Z}^3$ and a set of space–time replacement events at pairwise ℓ^∞ distance greater than d such that two trajectories, subject to the same replacement symbols, disagree on an entire plane at every time.

This statement is proved in Theorem 8.1. The automaton may depend on d.

The countdown lift applies to any cellular automaton. A countdown read from the next height gates each update of a data layer, the lifted rule is nilpotent, and under uniform replacement each data layer runs the original rule with independent faults. Applied to the computable family of our paper [7], whose ergodicity at a fixed noise rate is equivalent to halting, the lift gives the following.

Theorem C. There is a total computable map $M \mapsto N_M$ from Turing machines to radius-one cellular automata on $\mathbb{Z}^9$, all nilpotent within the same number of steps, with the following property. For every rate $\varepsilon \in [\varepsilon_0/2, \varepsilon_0]$, where $\varepsilon_0 = 490000^{-99}$, uniform replacement noise at rate ε makes N_M ergodic if and only if M halts, and if M does not halt it has two translation-invariant stationary laws.

This is Theorem 6.1, which also gives explicit ergodic ranges at small and large noise for every member. The alphabet of N_M grows with M, as in [7]; the rates, the radius, the dimension and the nilpotency time do not. So ergodicity at a fixed noise rate stays undecidable when the deterministic rules are restricted to nilpotent ones. Several countdowns in one rule give several windows.

Theorem D. For every $n \geq 1$, there is a radius-one nilpotent cellular automaton on $\mathbb{Z}^3$ whose uniform replacement perturbation has two translation-invariant stationary laws at every rate in n disjoint open intervals, separated in order by $n-1$ open intervals of uniform exponential ergodicity. It is also uniformly exponentially ergodic at all sufficiently small and all sufficiently large positive rates.

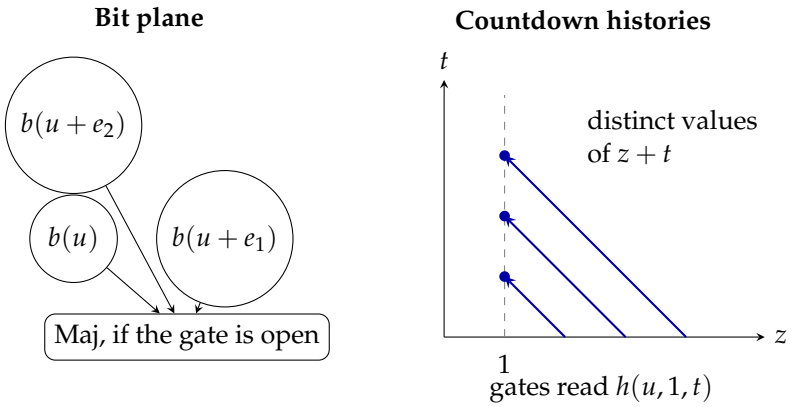


Figure 1: The bits use majority within one horizontal plane. The countdown read by each successive gate follows a different diagonal in the (z,t) coordinates. Independent randomness on these disjoint diagonals gives independent gates.

This is Theorem 7.1. For every n it gives a nilpotent automaton whose ergodicity changes at least $2n$ times as the noise rate varies, which bears on the question of Marsan, Sablik and Törmä [4] of which sets of noise rates can occur as ergodicity sets; it does not determine the status between the stated intervals. Whether one nilpotent rule can have infinitely many such windows is open (Remark 7.2).

These rules erase a configuration only after a long countdown; the rule of Theorem A needs $2^{88}+1$ steps. A rule that erases everything in one step is ergodic under every zero-range noise, since after one noisy step the configuration has a fixed product law (Proposition 9.1). Two steps suffice.

Theorem E. There is a cellular automaton F on $\mathbb{Z}^3$ with four symbols such that $F \circ F$ is constant and whose uniform replacement perturbation has two translation-invariant stationary laws at every rate in $[2^{-44}, 2^{-42}]$. The same rule is uniformly exponentially ergodic at every rate in $(0, 1/(3 \cdot 2^{50}))$ and in $(2/3, 1]$.

This is Theorem 9.4. Each site reads a seed, the word of 2^{51} auxiliary symbols above it, and every deterministic update resets the auxiliary symbols to zero; the radius of the rule is therefore 2^{51}. Theorems A and E trade nilpotency time against radius. Exact rational bins for the seed turn this mechanism into a compiler.

Theorem F. Fix $d \geq 1$ and rationals $0 < \varepsilon < 2\rho \leq 1$. There is a total computable map $P \mapsto F_P$ from attractive binary probabilistic cellular automata on $\mathbb{Z}^d$ with rational transition probabilities in $[\rho, 1-\rho]$ to cellular automata on $\mathbb{Z}^{d+1}$ with $F_P \circ F_P$ constant, with the following property. Uniform replacement at rate ε after F_P is ergodic, is uniformly exponentially ergodic, or has two translation-invariant stationary laws, if and only if P has the same property.

This is Theorem 9.7, and every horizontal plane of the compiled system runs P exactly from time 1 (Theorem 9.7(ii)). The rate and the dimension are fixed before P is given; the alphabet and the radius depend on P. Compiling noisy north-east-centre majority gives a second two-step example, whose planes are exactly the noisy majority process (Corollary 9.9). The same compiler, applied to a prescribed law of local maps, turns the family of [7] into a family on $\mathbb{Z}^9$, nilpotent in two steps, whose ergodicity at one rate is equivalent to halting (Remark 9.10). That family is incomparable with Theorem C: it is nilpotent in two steps, but it loses radius one and the interval of rates.

Toom proved stability of north-east-centre majority under small errors [6]. We use the

Result	Behavior under noise	Deterministic rule
Marcovici–Sablik–Taati [2]	Ergodicity under all sufficiently small perturbations of each fixed nilpotent rule.	Nilpotent
Marsan–Sablik–Törmä [4]	One-dimensional example with ergodicity at low and high noise, and non-ergodicity between.	Cyclic clock
Douglas–Mghirbi [7]	Computable family on $\mathbb{Z}^8$ whose ergodicity at each fixed small rate is equivalent to halting.	Quorum rule
This paper, Theorems A and D	Three-dimensional example with the same three regimes under uniform replacement noise; for each n, an example with n separated non-ergodic windows.	Nilpotent
This paper, Theorem C	Computable family on $\mathbb{Z}^9$ whose ergodicity at each rate of a fixed interval is equivalent to halting.	Nilpotent
This paper, Theorems E and F	Four-symbol example on $\mathbb{Z}^3$ with two translation-invariant stationary laws on an interval of rates; exact transfer of ergodicity, exponential ergodicity and coexistence from attractive binary PCA in one more dimension.	Two-step nilpotent

Table 1: The nilpotent question, the earlier two-transition phenomenon, undecidable ergodicity at a fixed rate and the least nilpotency time.

explicit infinite-plane estimate in Gács's exposition [1]. The extra spatial coordinate in our construction lets the countdowns move toward a fixed data plane without reusing their random histories there; Figure 1 illustrates this independence.

Marsan, Sablik and Törmä already established the phenomenon of ergodicity at low and high noise, with non-ergodicity in between, for a one-dimensional cellular automaton [4]. Their rule retains a cyclic clock in the absence of noise. Our contribution is a nilpotent rule with this behavior, and an exact reduction to a familiar two-dimensional majority process. The dependence of stationary laws on vanishing noise for nilpotent automata is also studied by Marsan and Sablik [3].

Reading guide. Sections 2 and 3 define the model and derive its independent gates. Section 4 imports the majority-stability estimate and proves coexistence of stationary laws. Section 5 gives the coupling proof for the two ergodic regimes. Section 6 lifts the undecidable family of [7] to a uniformly nilpotent family on $\mathbb{Z}^9$, and Section 7 gives, for each n, a nilpotent rule with n separated memory windows. Section 8 proves the separated-event statement directly. Section 9 shows that one deterministic step never suffices and two do: it proves Theorem E for a four-symbol seeded rule and gives the seed compiler of Theorem F. The non-elementary inputs are Toom's stability theorem in the stated quantitative form and, in Section 6 and Remark 9.10 only, the machine-to-tiling compiler, the three-token theorem and the Common local maps lemma of [7].

2 The rule and the noise

Write $\mathbb{N} = \{0, 1, \dots\}$. Fix an integer $H \geq 1$ and write a site of $\mathbb{Z}^3$ as (u, z) with $u \in \mathbb{Z}^2$ and $z \in \mathbb{Z}$. Its state belongs to

$$\mathcal{A}_H = \{0, 1\} \times \{0, 1, \dots, H\}.$$

The two coordinates of a configuration are denoted by b and h. Let e_1, e_2 be the positive coordinate vectors in $\mathbb{Z}^2$, and let $a_+ = \max(a, 0)$. Define the cellular automaton F_H by

$$(F_H h)(u, z) = (h(u, z+1) - 1)_+, \tag{1}$$

$$(F_H b)(u, z) = \mathbf{1}_{\{h(u,z+1)>0\}} \operatorname{Maj}\big(b(u, z), b(u + e_1, z), b(u + e_2, z)\big). \tag{2}$$

The notation $F_H h, F_H b$ refers to the coordinates of $F_H(b, h)$. Its neighborhood consists of 0, $(e_1, 0)$, $(e_2, 0)$ and $e_3 = (0, 0, 1)$. Thus its radius in the ℓ^∞ metric is one.

Proposition 2.1. *For every configuration, $F_H^{H+1}(b, h)$ is the uniform state $(0, 0)$. The least such common exponent is $H + 1$.*

Proof. Induction in (1) gives

$$h_t(u, z) = (h_0(u, z + t) - t)_+.$$

Consequently all countdowns are zero at time H. The next bit update is zero at every site, proving the first assertion. Starting with $b_0 \equiv 1$ and $h_0 \equiv H$ gives $h_t \equiv H - t$ and $b_t \equiv 1$ for $0 \le t \le H$. This configuration has not reached $(0, 0)$ at time H, proving minimality. □

For $0 < \varepsilon \le 1$, the noisy rule $P_{H,\varepsilon}$ first applies F_H and then, independently at each site and time, replaces the whole symbol with probability ε by a uniform element of $\mathcal{A}_H$. More explicitly, at times $t \ge 1$ let

$$R_t(u, z) \sim \text{Bernoulli}(\varepsilon), \qquad V_t(u, z) \sim \text{Bernoulli}(1/2), \qquad U_t(u, z) \sim \text{Unif}\{0, \dots, H\}$$

be mutually independent over all indices. If $R_t = 1$, set $(b_t, h_t) = (V_t, U_t)$ at that site; otherwise use (1)–(2) from time $t - 1$. Every one-site transition probability is at least $\varepsilon/[2(H + 1)]$. The noise is therefore positive and has zero range in the terminology of [2].

Let $\Omega_H = \mathcal{A}_H^{\mathbb{Z}^3}$ with the product topology. A stationary law is a probability measure μ satisfying $\mu P_{H,\varepsilon} = \mu$. We call the PCA *ergodic* if it has a unique stationary law and the law from every initial configuration converges weakly to it. It is *uniformly exponentially ergodic* if, for some $C, c > 0$ and every finite $W \subset \mathbb{Z}^3$,

$$\sup_{x \in \Omega_H} d_{\mathrm{TV}}\big((\delta_x P_{H,\varepsilon}^t)_W, \mu_W\big) \le C|W|e^{-ct}. \tag{3}$$

All constants here may depend on H and ε.

The space Ω_H is compact and metrizable. Since a local observable after one update depends continuously on finitely many input symbols, the transition operator is Feller. In particular a stationary law exists: any weak limit point of the Cesàro averages $T^{-1} \sum_{t=0}^{T-1} \nu P_{H,\varepsilon}^t$ is stationary. Indeed, applying $P_{H,\varepsilon}$ to this average changes its integral against a bounded continuous observable by at most twice the supremum norm divided by T. We will use this elementary compactness argument for both existence and coexistence.

3 Countdowns and independent gates

Write $r = 1 - \varepsilon$ in this section. At a site (u, z) and time t, the countdown reads its past along

$$(u, z + a, t - a), \qquad a = 0, 1, \dots, t. \tag{4}$$

Along this diagonal the value of $z + t$ is constant. If the most recent replacement on it occurred a steps ago, the present countdown is $(U - a)_+$. In the absence of such a replacement it is $(h_0(u, z + t) - t)_+$. Only the last H update positions can affect the present value: a reset of age at least H has already reached zero, as has every possible initial countdown after H steps.

Lemma 3.1. *The countdown process has a stationary product law $\pi_{H,\varepsilon}^{\otimes \mathbb{Z}^3}$. Its one-site marginal is the law of $(U - A)_+$, where U is uniform on $\{0, \dots, H\}$ and A is independent with*

$$\mathbb{P}(A = a) = \varepsilon r^a, \qquad a \in \mathbb{N}.$$

For every $t \geq H$ the countdown configuration has this product law from every initial configuration, and it is independent of the initial countdowns.

Proof. For one diagonal, the geometric variable A records the age of the last replacement in an indefinitely extended past. At one further step its age is zero with probability ε, and otherwise increases by one, which preserves this geometric law. The reset mark U is independent and uniform, so the stated marginal is invariant under the countdown transition.

At a fixed time, distinct sites follow disjoint diagonals in (4). Product initial countdowns and independent replacement marks therefore give independent countdowns again. This proves stationarity of the product law. At time $t \geq H$, truncating each diagonal to ages $0, \dots, H-1$ gives exactly the same value: if there has been no replacement in these H positions, the value is zero. This is the same function of the same finite set of marks for every initial countdown configuration. Disjointness of the diagonals proves the final assertions. □

Fix a horizontal bit plane, say $z = 0$, and define its gate by

$$G_t(u) = \mathbf{1}_{\{h_t(u,1)>0\}}, \qquad t \geq 0.$$

The gate is read in the update of $b_{t+1}(u, 0)$. Initialize the countdowns with the product law in Lemma 3.1, independently of all later marks, and initialize the bits with any deterministic configuration.

Lemma 3.2 (Exact gate law)**.** *The variables $G_t(u)$ are independent over $(u, t) \in \mathbb{Z}^2 \times \mathbb{N}$. They are independent of every replacement variable $R_s(v, 0), V_s(v, 0)$ in the bit plane. Their common closed-gate probability is*

$$q_H(\varepsilon) = \mathbb{P}(G_t(u) = 0) = \frac{1}{H+1} \sum_{j=0}^{H} (1-\varepsilon)^j = \frac{1 - (1-\varepsilon)^{H+1}}{\varepsilon(H+1)}. \tag{5}$$

From arbitrary deterministic initial countdowns, the same assertions hold for the gates with $t \geq H$.

Proof. The primitive variables determining $G_t(u)$ lie on the diagonal

$$(u, 1 + a, t - a), \qquad 0 \leq a < t,$$

together with the initial height at $(u, 1 + t)$ when it is needed. A second gate either has a different u or a different value of $z + t = 1 + t$. Its determining variables are then disjoint from the first gate's. All initial countdowns are independent, so the gates are independent. Every determining site has $z \geq 1$, while the displayed bit-plane replacement variables have $z = 0$. This proves their independence as well.

By Lemma 3.1, a gate is closed exactly when $U \leq A$. Averaging first over U gives

$$\mathbb{P}(U \leq A) = \frac{1}{H+1} \sum_{j=0}^{H} \mathbb{P}(A \geq j) = \frac{1}{H+1} \sum_{j=0}^{H} r^j.$$

Summing this geometric series proves (5). For $t \geq H$, the gate depends only on the last H update positions, regardless of its initial countdown, so the preceding independence argument applies verbatim. □

The qualification about the initial countdowns is necessary. For example, from $h_0 \equiv H$, the gates at time zero are deterministically open. The stationary initialization gives the exact law (5) from the first bit update; arbitrary initialization gives it after the finite countdown transient.

Proposition 3.3 (The projected majority process)**.** *Under the stationary countdown initialization, $B_t(u) = b_t(u, 0)$ is a binary PCA with independent local random choices and update*

$$B_{t+1}(u) = \begin{cases} V_{t+1}(u, 0), & R_{t+1}(u, 0) = 1, \\ G_t(u)\,\mathrm{Maj}(B_t(u), B_t(u + e_1), B_t(u + e_2)), & R_{t+1}(u, 0) = 0. \end{cases} \tag{6}$$

The indicators

$$I_{t+1}(u) = \mathbf{1}_{\{R_{t+1}(u,0)=1 \text{ or } G_t(u)=0\}} \tag{7}$$

are independent Bernoulli variables of parameter

$$p_H(\varepsilon) = \varepsilon + (1 - \varepsilon) q_H(\varepsilon). \tag{8}$$

At every update where $I_{t+1}(u) = 0$, the bit obeys north-east-centre majority exactly.

Proof. Equation (6) is the bit coordinate of the original update. Its right-hand side uses the plane's bits, its past gate variables, and its plane replacement variables. By Lemma 3.2, the new gates and replacements are independent of all these earlier variables and independent across u. This proves the PCA assertion and (8). A zero indicator in (7) means an open gate and no replacement, giving the last assertion. □

For clarity, the independent indicators describe opportunities for an error; a replacement may happen to write the majority value. The exact transition probability of the projected PCA is

$$\mathbb{P}(B_{t+1}(u) = 1 \mid B_t) = \frac{\varepsilon}{2} + (1 - \varepsilon)(1 - q_H(\varepsilon))\,\mathrm{Maj}(B_t(u), B_t(u + e_1), B_t(u + e_2)). \tag{9}$$

The actual error probability is $\varepsilon/2$ when the majority is zero and $\varepsilon/2 + (1 - \varepsilon) q_H(\varepsilon)$ when it is one. Both are bounded by $p_H(\varepsilon)$.

4 Memory at a positive noise rate

We state precisely the stability estimate needed here, taken from Gács's exposition [1] of the Berman–Simon proof [5]. The parameter ρ below bounds local update errors relative to majority, and is distinct from the replacement probability ε of our model.

Theorem 4.1 (Toom stability, quantitative specialization)**.** *Let a binary process on $\mathbb{Z}^2$, started from the constant bit $a \in \{0, 1\}$, be a ρ-perturbation of north-east-centre majority: conditional on the past, its site updates are independent and each disagrees with majority with probability at most ρ. If*

$$\rho < \tau := \frac{1}{32 \cdot 24^8}, \tag{10}$$

then at every site and time its probability of disagreeing with a is at most 96ρ.

This is the infinite-plane case $m = \infty$ of the main theorem of [1],[1] specialized as follows. Take the auxiliary one-dimensional space in its Definition 1.5 to be a singleton and its transition

[1] In the displayed perturbation condition of [1], the comparison with the prescribed output is printed as equality. We use the disagreement event specified by its subsequent definition of Noise and its proof.

rule to be the identity. The resulting three-dimensional rule is exactly NEC majority on the remaining two coordinates. The constant reference trajectory is a at every site and time. The constants stated there are $r = 24$, threshold $(32r^8)^{-1}$, and error bound $4r\rho$. Independent site updates with the displayed conditional error bound satisfy its perturbation hypothesis, including conditioning on the other simultaneous outputs. Outside Section 6 and Remark 9.10, this is the sole imported estimate; Toom's original theorem is [6]. A one-point auxiliary space is admitted by the definitions of [1]; the neighbours $n \pm 1$ then coincide with n, and its perturbation condition is read for the three-dimensional rule as in the theorem.

Theorem 4.2. *If $p_H(\varepsilon) < \tau$, then $P_{H,\varepsilon}$ has two translation-invariant stationary laws μ^0, μ^1 satisfying, at every site $v \in \mathbb{Z}^3$,*

$$\mu^0\{b(v) = 1\} \le 96 p_H(\varepsilon), \qquad \mu^1\{b(v) = 1\} \ge 1 - 96 p_H(\varepsilon). \tag{11}$$

In particular it is non-ergodic. The condition holds for

$$\varepsilon_* = 2^{-44}, \qquad H_* = 2^{88}. \tag{12}$$

Proof. For $a = 0, 1$, let ν^a be the initial law with every bit equal to a and countdowns drawn from the stationary product law. For every horizontal plane, Proposition 3.3 is a $p_H(\varepsilon)$-perturbation in the sense of Theorem 4.1. Its error probability conditional on the entire plane past is bounded by (8), and updates at the current time are conditionally independent. Thus

$$\mathbb{P}_{\nu^a}\{b_t(v) \ne a\} \le 96 p_H(\varepsilon) \qquad (v \in \mathbb{Z}^3,\ t \ge 0).$$

The planes need not be independent; the estimate uses their individual marginal processes.

Take a weak limit point μ^a of the Cesàro averages of $\nu^a P_{H,\varepsilon}^t$. The Feller argument in Section 2 proves stationarity. The initial laws are translation invariant and the transition commutes with translations, so the limit laws are also translation invariant. The one-site events in the preceding display are clopen cylinders, and their bounds pass to the limit, proving (11). Since $192\tau < 1$, the two bounds are disjoint and the laws differ.

For the explicit choice, (5) gives

$$q_{H_*}(\varepsilon_*) \le \frac{1}{\varepsilon_*(H_* + 1)} < 2^{-44}, \qquad p_{H_*}(\varepsilon_*) < 2^{-43}.$$

Also $32 \cdot 24^8 = 2^{29} 3^8 < 2^{42}$ because $3^8 = 6561 < 8192$. Hence $2^{-43} < \tau$, as required. □

More generally, for every fixed $0 < \varepsilon < \tau$, a sufficiently large H satisfies the hypothesis, since $q_H(\varepsilon) \le 1/[\varepsilon(H + 1)]$. Thus the counterexample can be placed at arbitrarily small positive replacement rates by changing the automaton. For any one chosen H, sufficiently small noise has a different behavior, which we now prove. The existence of some H and ε with two stationary laws needs only Toom's theorem [6] with an unspecified threshold; the constants of [1] enter only in the explicit choice above.

5 Ergodicity at small and large noise

Couple two trajectories from arbitrary configurations by using the same replacement indicators and the same replacement symbols. Their countdowns agree everywhere by time H, even if no replacement occurs on a particular diagonal. Once the countdowns agree, a closed gate or a replacement makes the corresponding bit outputs agree. An open gate without a

replacement can pass a disagreement only from one of the three majority parents in its own plane.

Put

$$\alpha_H(\varepsilon) = (1-\varepsilon)(1-q_H(\varepsilon)), \qquad \lambda_H(\varepsilon) = 3\alpha_H(\varepsilon). \tag{13}$$

The first quantity is the probability that one bit update can depend on its preceding bits.

Theorem 5.1. *If $\lambda_H(\varepsilon) < 1$, then $P_{H,\varepsilon}$ has a unique stationary law μ and, for every finite $W \subset \mathbb{Z}^3$ and $n \geq 1$,*

$$\sup_{x \in \Omega_H} d_{\mathrm{TV}}\big((\delta_x P_{H,\varepsilon}^{H+n})_W, \mu_W\big) \leq |W| \lambda_H(\varepsilon)^n. \tag{14}$$

It is uniformly exponentially ergodic whenever

$$0 < \varepsilon < \frac{2}{3H} \qquad \text{or} \qquad \frac{2}{3} < \varepsilon \leq 1. \tag{15}$$

The qualitative statements are known: ergodicity under small noise is [2, Theorem 3.9], and ergodicity under large noise holds for every perturbed cellular automaton. The explicit ranges and the coupling bound above are what we add.

Proof. Fix a site $v = (u, z)$ and suppose its bits in the two coupled trajectories differ at time $H + n$. Trace backward from this disagreement. At each step there must be an open gate, no replacement at the output, and at least one disagreeing majority parent. Consequently there is a path of length n ending at time H with each backward spatial displacement in $\{0, e_1, e_2\}$, all within the plane at height z. Every update on this path is open and unreplaced.

There are 3^n possible paths. For a fixed path, the relevant gates are those with input times $H, H+1, \ldots, H+n-1$ in one fixed plane. By the last assertion of Lemma 3.2, they are independent with open probability $1 - q_H(\varepsilon)$. They are independent of all replacements in that plane. The probability that every update on the path is open and unreplaced is therefore exactly $\alpha_H(\varepsilon)^n$. The union bound gives

$$\mathbb{P}\{b_{H+n}(v) \neq b'_{H+n}(v)\} \leq \lambda_H(\varepsilon)^n. \tag{16}$$

This argument does not require any independence between the initial disagreements and these gates: it bounds the event that a path of open, unreplaced updates exists. The countdowns already agree. Summing (16) over W bounds the probability of any full-symbol disagreement in W, uniformly over the two initial configurations.

A stationary law exists by compactness. Couple a deterministic initial configuration to a sample from that law and integrate the preceding uniform bound. The coupling inequality gives (14), hence convergence to this stationary law from every start. Coupling samples from any two stationary laws gives the same bound at every n; sending n to infinity proves uniqueness. If $0 < \lambda_H(\varepsilon) < 1$, take $c = -\log \lambda_H(\varepsilon)$ and enlarge the prefactor to cover $t \leq H$. If $\lambda_H(\varepsilon) = 0$, the trajectories agree after at most $H + 1$ steps, which also implies (3).

It remains to check (15). Bernoulli's inequality in (5) yields

$$q_H(\varepsilon) \geq \frac{1}{H+1} \sum_{j=0}^{H} (1 - j\varepsilon) = 1 - \frac{H\varepsilon}{2}.$$

Thus $\lambda_H(\varepsilon) \leq 3H\varepsilon/2 < 1$ in the small-noise range. For large noise, $\lambda_H(\varepsilon) \leq 3(1-\varepsilon) < 1$. □

Corollary 5.2 (Re-entrance)**.** *For the single rule F_{H_*} in (12), there is an open interval of positive noise rates containing ε_* on which the PCA is non-ergodic. It is uniformly exponentially ergodic at every rate in $(0, 2/(3H_*))$ and $(2/3, 1]$.*

Proof. For fixed H_* the expression $p_{H_*}(\varepsilon)$ is continuous on $(0,1]$. The strict inequality proved in Theorem 4.2 persists in a neighborhood of ε_*. The two ergodic ranges follow from Theorem 5.1. They lie on either side of ε_* because $2/(3 \cdot 2^{88}) < 2^{-44} < 2/3$. □

These sufficient conditions establish ergodicity, non-ergodicity and ergodicity in that order as the noise rate increases. They do not locate the boundaries between the regimes. The exact contraction condition $3(1-\varepsilon)(1-q_H(\varepsilon)) < 1$ can also apply beyond the two intervals in (15).

6 Nilpotency and undecidable noisy ergodicity

The countdown mechanism transfers an undecidable noisy-ergodicity classification to a family consisting entirely of nilpotent automata. The dimension is nine and the countdown bound is the same for every member.

Published input and credit. We use our paper [7], hereafter paper A, specifically [7, Corollary 9.21 and Lemma 8.11]; the latter is its Common local maps lemma. The machine-to-Wang-set compiler and the quorum and contour estimates are imported from that paper. So is its three-token theorem, whose proof includes one finite lemma that is checked by computer there [7, Lemma 9.12]. The lift, its burn-in justification, the endpoint bounds and the multiple-window construction are proved here. The countdown mechanism and diagonal independence are those of the preceding sections. We assume no independence between fuel tracks or between different horizontal planes.

Set

$$\varepsilon_0 = 490000^{-99}, \quad H_0 = 2 \cdot 490000^{198}, \quad I_0 = [\varepsilon_0/2, \varepsilon_0], \quad \varepsilon_+ = \varepsilon_0/2. \tag{17}$$

The rate ε_0 is the rate of [7, Corollary 9.21].

Theorem 6.1 (Nilpotency and undecidability)**.** *There is a total computable map $M \mapsto N_M$ from Turing machines to finite-alphabet, radius-one cellular automata on $\mathbb{Z}^9$ such that:*

(i) *every N_M maps every configuration to the same uniform fixed configuration after $H_0 + 1$ deterministic updates;*

(ii) *for every $\varepsilon \in I_0$, uniform replacement on the entire alphabet of N_M at rate ε is ergodic if and only if M halts;*

(iii) *for every M, this perturbation is uniformly exponentially ergodic whenever*

$$0 < \varepsilon < \frac{2}{63H_0} \quad \text{or} \quad \frac{62}{63} < \varepsilon \le 1.$$

The alphabet of N_M depends on M; the rate interval, radius, dimension and nilpotency time do not. If M does not halt and $\varepsilon \in I_0$, there are at least two translation-invariant stationary laws. Thus ergodicity at the single rational rate ε_+ is undecidable even within this uniformly nilpotent family.

6.1 The exact imported interface

Paper A gives a total computable map $M \mapsto R_M$ from Turing machines to finite Wang sets [7, Definition 6.17]. Let T_{R_M} be the set of legal 3×3 blocks of R_M and put

$$\mathcal{B}_M = (T_{R_M} \times \{0,1\}) \cup \{\perp\}.$$

The symbols other than $\perp$ are *live*. The planar checker keeps the symbol at a site precisely when that symbol is live and its eight Moore neighbors are live and compatible with it, that is, they carry the same checker bit and their blocks agree with its block where they overlap; otherwise it returns $\perp$ [7, Definition 2.5]. In $\mathbb{Z}^2 \times \mathbb{Z}^6$, perform this checker in each $\mathbb{Z}^2$ plane. The quorum rule $F_M = F_{R_M}^{[6]}$ returns a live symbol if at least six of the seven checked readings at transverse displacements $0, e_1, \dots, e_6$ equal it, and returns $\perp$ otherwise [7, Definition 2.7]. Its neighborhood is

$$\mathcal{N} = \{-1, 0, 1\}^2 \times \{0, e_1, \dots, e_6\}, \qquad |\mathcal{N}| = 63. \tag{18}$$

All finite tables are enumerable from M. If M does not halt, R_M tiles [7, Proposition 6.18]; a *cloned tiling* is then the configuration that carries the blocks of one tiling, with checker bit 0, identically on every plane [7, Definition 4.1]. If M halts, R_M is refuted by three tokens [7, Theorem 9.1], with some finite refutation time τ_M.

A random field of local maps is γ-bounded [7, Definition 8.10] if the maps agree with F_M away from designated faults and, for every finite set S of logical spacetime points of positive time,

$$\mathbb{P}(\text{every point of } S \text{ is faulty}) \le \gamma^{|S|}. \tag{19}$$

Logical time is counted from the start of the field: from physical time 0, or from physical time H_0 after the burn-in below. The same maps drive both starts. Paper A's Common local maps lemma [7, Lemma 8.11] says the following. For $0 < \gamma \le 2\varepsilon_0$, write

$$D_0 = 350, \qquad \beta = 2D_0^2\gamma^{1/99}, \qquad \ell = 2D_0^2\gamma^{1/49},$$

with the constants of [7, Corollary 9.21]. If M halts, the common-map coupling from any two deterministic starts obeys

$$\mathbb{P}(X_s(v) \ne Y_s(v)) \le \frac{2\gamma^{-\tau_M/99}\beta^s}{1-\beta}. \tag{20}$$

If M does not halt, a run from any cloned tiling ξ differs from ξ at a given site with probability at most $\gamma(1 + 2\ell/(1-\ell)) < 1.01\gamma$; a run from bottom is live there with probability less than 1.01γ, at every time. These conclusions allow arbitrary faulty maps satisfying (19). We will have iid faults and constant maps at faults.

For completeness, the numerical margins are explicit:

$$\beta \le 2^{1/99}/2 < 13/25, \qquad \ell \le \frac{2^{1/49}}{8 \cdot 350^2} < \frac{1}{490000}.$$

Indeed $(26/25)^{99} \ge 1 + 99/25 > 2$; also $99/49 > 2$ and $2^{1/49} < 2$. Thus $1 + 2\ell/(1-\ell) < 1.01$. Since $\gamma < 1/1000$, the gap $1 - 2.02\gamma$ is positive. Here τ_M is the time coordinate of any finite refutation witness. It is unnecessary for constructing N_M; no total computable bound on this convergence parameter is asserted.

6.2 A general countdown lift

Let F be any cellular automaton on $\mathcal{B}^{\mathbb{Z}^d}$ whose local rule f reads $m \ge 1$ inputs, and choose $\perp \in \mathcal{B}$. For an integer $K \ge 1$, its lift on $\mathbb{Z}^d \times \mathbb{Z}$ has alphabet $\mathcal{B} \times \{0, \dots, K\}$ and rule

$$h'(v, z) = (h(v, z+1) - 1)_+, \tag{21}$$

$$b'(v, z) = \begin{cases} F(b(\cdot, z))(v), & h(v, z+1) > 0, \\ \perp, & h(v, z+1) = 0. \end{cases} \tag{22}$$

The data symbol b need not be binary. Quiescence of $\perp$ for F is unnecessary.

Lemma 6.2 (Exact slice law and burn-in)**.** *The lift is nilpotent by time $K+1$. Under uniform replacement of its entire symbol at rate $0 < \varepsilon \leq 1$, put*

$$q_K(\varepsilon) = \frac{1}{K+1}\sum_{j=0}^{K}(1-\varepsilon)^j = \frac{1-(1-\varepsilon)^{K+1}}{\varepsilon(K+1)}, \qquad p_K(\varepsilon) = \varepsilon + (1-\varepsilon)q_K(\varepsilon).$$

For any fixed height z, from time K its data evolve by independent local maps with iid fault indicators of parameter $p_K(\varepsilon)$. At a nonfault point the map is F. The entire future slice map field is independent of the data slice at time K, and of the pair of data slices there in a common-noise coupling. With stationary product fuel initialization, independent of the initial data and all subsequent noise, the exact slice law holds from time zero. Its kernel is

$$Q(s \mid y) = (1-\varepsilon)(1-q_K(\varepsilon))\mathbf{1}_{\{s=f(y)\}} + (1-\varepsilon)q_K(\varepsilon)\mathbf{1}_{\{s=\perp\}} + \frac{\varepsilon}{|\mathcal{B}|}, \tag{23}$$

where $y \in \mathcal{B}^m$ is the tuple of inputs read by the local rule f of F.

Proof. Noiselessly, $h_t(v,z) = (h_0(v,z+t)-t)_+$. All fuels vanish at K and the next data update is bottom. The resulting uniform state is fixed because its gates are closed.

At each physical spacetime point use independent variables: a Bernoulli coin R with parameter ε, a uniform fuel mark $U \in \{0,\dots,K\}$, and a uniform data mark $V \in \mathcal{B}$. On $R=1$ write (V,U); otherwise apply the deterministic rule. The two coordinates share R, exactly as uniform replacement requires.

The history of the gate reading $h_t(v,z+1)$ lies on

$$(v, z+1+a, t-a), \quad a \geq 0,$$

with invariant $(v, z+1+t)$. A reset of age a supplies $(U-a)_+$. Resets of age at least K, and all initial fuels after K updates, supply zero. Thus all dependence on initial fuel vanishes by time K. The stationary marginal is $(U-A)_+$ where $\mathbb{P}(A=a) = \varepsilon(1-\varepsilon)^a$ independently of U. Independent copies across sites are stationary, since distinct sites at one time use disjoint characteristics. Averaging over U gives

$$\mathbb{P}((U-A)_+ = 0) = \frac{1}{K+1}\sum_{j=0}^{K}\mathbb{P}(A \geq j) = q_K(\varepsilon).$$

At the fixed data height z, different logical (v,t) gates have different invariants or different v, hence disjoint histories. All determining sites are strictly above z, disjoint from every replacement primitive in the data slice. The gates are therefore independent over logical spacetime and independent of in-slice replacements. With stationary product fuels this holds from time zero. For arbitrary initial fuels, gates of input times $t \geq K$ have forgotten the initial state and have the same iid law. The union of a closed gate and an in-slice replacement is an iid fault with parameter $p_K(\varepsilon)$. A replacement writes V; a closed gate without replacement writes bottom; all other points apply F. This proves (23).

It remains to justify restarting at a random time-K slice. Its history through K uses only initial data and fuels, in-slice replacements through K, and gates of input times $0,\dots,K-1$, whose invariants are $z+1,\dots,z+K$. Future gates have invariants $z+K+1, z+K+2,\dots$, use only noise, and lie strictly above the plane. Future in-slice replacements are fresh. These sets of primitives are disjoint, proving independence from the pair of time-K data slices. The argument also permits random initial configurations independent of the noise. Independence from other slices' past is not asserted or required. □

Lemma 6.3 (Endpoint coupling). *Put $\lambda_K(\varepsilon) = m(1-\varepsilon)(1-q_K(\varepsilon))$. If $\lambda_K(\varepsilon) < 1$, the lift has a unique stationary law μ and for every finite physical window W and integer $s \geq 1$,*

$$\sup_x d_{\mathrm{TV}}((\delta_x P^{K+s})_W, \mu_W) \leq |W|\lambda_K(\varepsilon)^s. \tag{24}$$

This holds in particular for $0 < \varepsilon < 2/(mK)$ or $1 - 1/m < \varepsilon \leq 1$.

Proof. Use common physical noise for two arbitrary starts. Their fuels agree everywhere by K. Any later data discrepancy needs an open unreplaced update and a discrepant parent among the m inputs in its slice. Trace s steps backward from a discrepancy at $K+s$. There are at most m^s possible paths. For any fixed path its distinct spacetime gates and replacement coins are independent, so all updates are open and unreplaced with probability $[(1-\varepsilon)(1-q_K(\varepsilon))]^s$. The union bound gives $\lambda_K(\varepsilon)^s$. This bounds existence of a transmitting path and does not condition on the data at K.

Sum over W to bound full-symbol discrepancies. A stationary law exists by compactness and Feller continuity. Coupling to it proves (24); coupling two stationary laws proves uniqueness by letting $s \to \infty$. For $0 < \lambda_K < 1$, one may take exponential constants $c = -\log \lambda_K$ and $C = \lambda_K^{-K}$ for all times. If $\lambda_K = 0$, there is finite-time coupling.

Finally, $(1-\varepsilon)^j \geq 1 - j\varepsilon$ implies $1 - q_K(\varepsilon) \leq \varepsilon K/2$. Hence $\lambda_K \leq m\varepsilon K/2$ and $\lambda_K \leq m(1-\varepsilon)$, giving both intervals. At $\varepsilon = 1$ one update is iid uniform. □

Proof of Theorem 6.1. Use $F = F_M$, $d = 8$, $K = H_0$ in the lift. Its alphabet is $\mathcal{D}_M = \mathcal{B}_M \times \{0, \ldots, H_0\}$. Its neighborhood consists of $\mathcal{N} \times \{0\}$ and the positive ninth-coordinate neighbor: 64 sites, of radius one. Its table is total computable from M. The slice lemma gives nilpotency by $H_0 + 1$, uniformly in M.

For $\varepsilon \in I_0$,

$$q_{H_0}(\varepsilon) \leq \frac{1}{\varepsilon(H_0+1)} \leq \frac{2}{\varepsilon_0(H_0+1)} < \varepsilon_0, \qquad p_{H_0}(\varepsilon) < 2\varepsilon_0.$$

Fix $\gamma = 2\varepsilon_0$. For every finite set S of logical spacetime points of positive time within one slice, the exact identity

$$\mathbb{P}(S \text{ all faulty}) = p_{H_0}(\varepsilon)^{|S|} \leq \gamma^{|S|}$$

verifies the full joint hypothesis of paper A.

If M halts, couple two complete initial configurations using common physical noise. Fuels have coupled by H_0. By the burn-in independence in the slice lemma, condition on the pair of data slices at H_0 and apply (20). (Alternatively, the certificate event in the proof of [7, Lemma 8.11(ii)] depends on the fault field alone and contains the disagreement for every pair of starts, so no conditioning is needed.) For every finite physical window W and $s \geq 1$,

$$\sup_{x,y} \mathbb{P}(X_{H_0+s}|_W \neq Y_{H_0+s}|_W) \leq \frac{2|W|\gamma^{-\tau_M/99}\beta^s}{1-\beta}, \qquad \beta = 2^{1/99}/2 < 13/25. \tag{25}$$

This union bound uses one common full-process coupling; correlations between heights cause no difficulty. Coupling to a stationary law gives attraction and uniqueness. Uniform exponential constants, including the burn-in, are $c = -\log \beta$ and $C = 2\gamma^{-\tau_M/99}\beta^{-H_0}/(1-\beta)$. Every convergence constant is explicit for a finite refutation witness; that witness is unnecessary in the construction.

If M does not halt, clone a tiling configuration through all heights, and independently use stationary product fuels. For the other initial law use all-bottom data and stationary fuels.

Every slice has its iid map law from the first update. The imported memory estimates give, at every physical site and time,

$$\mathbb{P}_{\text{live}}(b_t \neq \perp) \geq 1 - 1.01\gamma, \qquad \mathbb{P}_{\text{bottom}}(b_t \neq \perp) \leq 1.01\gamma.$$

Take simultaneous subsequential Cesàro limits of these two full-process laws. Both limits are stationary by Feller continuity. The live-symbol cylinder is clopen, so the positive probability gap passes to the limits. The stationary laws are distinct. For translation invariance, first average a tiling over expanding planar boxes and take a limit supported on tilings. The boundary-to-volume estimate makes this law shift invariant. Clone it through all transverse coordinates and heights. Both initial full laws, and thus their Cesàro limits, are then translation invariant; all estimates survive integration over tilings.

Finally the endpoint lemma with $m = 63$ gives the claimed explicit small- and large-noise ranges. □

A second imported interface. The exact projected law (23) is F followed by replacement at rate $p_K(\varepsilon)$ with law

$$\nu = \frac{\varepsilon \text{ Unif}(\mathcal{B}) + (1-\varepsilon)q_K(\varepsilon)\delta_\perp}{p_K(\varepsilon)}.$$

Thus the proof of paper A's other-replacement-law corollary [7, Corollary 8.6 and Remark 9.22], which gives the per-site common-noise coupling bound for ν-replacement with the same constants, can replace (20); it is applied in each slice after burn-in and summed over a window exactly as above, and its part (ii) replaces the memory estimates. Take $K = 8 \cdot 490000^{198}$ and $\varepsilon \in [\varepsilon_0/4, \varepsilon_0/2]$. Then $q_K(\varepsilon) < \varepsilon_0/2$ and $p_K(\varepsilon) < \varepsilon_0$, within its original threshold. This route does not require the extension to arbitrary faulty maps. Physical noise in either construction remains exactly uniform replacement of the entire symbol.

7 Arbitrarily many separated memory windows

Theorem 7.1 (Any prescribed number of windows)**.** *For each integer $n \geq 1$ there is a radius-one nilpotent cellular automaton on $\mathbb{Z}^3$ with n disjoint open intervals of replacement rates having two translation-invariant stationary laws, separated in order by $n-1$ open intervals of uniform exponential ergodicity. It is also uniformly exponentially ergodic at all sufficiently small and all sufficiently large positive rates. All capacities and intervals are explicit.*

Proof. Set

$$L = 2^{50}(n+1), \quad H_i = L^{2(2n-i+1)} \ (1 \leq i \leq 2n), \quad K = H_1 = L^{4n}.$$

Use alphabet $\{0,1\} \times \prod_{i=1}^{2n}\{0,\dots,H_i\}$. Let $\mathcal{G}(h)$ be the indicator that for some j, $h_{2j-1} > 0$ and $h_{2j} = 0$. The rule is

$$h_i'(u,z) = (h_i(u,z+1) - 1)_+,$$
$$b'(u,z) = \mathcal{G}(h(u,z+1))\,\text{Maj}(b(u,z), b(u+e_1,z), b(u+e_2,z)).$$

All fuels vanish at K, and all bits at $K+1$. This is the least nilpotency exponent: initial bits one, $h_1 = K$ and all other fuels zero keep all bits one through time K.

Under uniform replacement of the whole symbol, the reset coin is shared; conditional on replacement its bit and fuel marks are independent uniforms. The stationary joint fuel vector

at a site is $((U_i - A)_+)_{i=1}^{2n}$, where the U_i are independent uniforms and $\mathbb{P}(A = a) = \varepsilon(1-\varepsilon)^a$ is one common age. The tracks are generally dependent. Independent copies of this vector across sites are stationary; all initial fuels are forgotten by K. The slice lemma's disjoint-characteristic proof applies to this vector. Therefore, within each bit slice, gates are iid over spacetime and independent of in-plane replacements, from stationary initialization or after burn-in K. Writing $g(\varepsilon) = \mathbb{P}(\mathcal{G} = 1)$, the projected kernel is

$$\mathbb{P}(B_{t+1}(u) = 1 \mid B_t) = \varepsilon/2 + (1-\varepsilon)g(\varepsilon)\,\mathrm{Maj}(B_t(u), B_t(u+e_1), B_t(u+e_2)).$$

Its iid majority-fault opportunities have probability $p(\varepsilon) = \varepsilon + (1-\varepsilon)(1-g(\varepsilon))$. For any track of capacity κ the preceding calculation gives

$$\mathbb{P}(h = 0) \le \frac{1}{\varepsilon(\kappa+1)}, \qquad \mathbb{P}(h > 0) \le \varepsilon\kappa/2. \tag{26}$$

Define

$$w_j = L^{-[4(n-j)+3]}, \quad I_j = (w_j/2, 2w_j) \qquad (1 \le j \le n).$$

Pair j has long capacity $L^{4(n-j)+4}$ and short capacity $L^{4(n-j)+2}$. For $\varepsilon \in I_j$, a union bound using (26) yields

$$1 - g(\varepsilon) \le \mathbb{P}(h_{2j-1} = 0) + \mathbb{P}(h_{2j} > 0) < 2/L + 1/L = 3/L.$$

Also $\varepsilon < 2/L^3 \le 1/L$, so $p(\varepsilon) < 4/L$. Since $L \ge 2^{51}$,

$$p(\varepsilon) < 2^{-49} < 2^{-42} < \tau,$$

with τ the threshold in (10). Indeed $32 \cdot 24^8 = 2^{29}3^8 < 2^{42}$. From constant bits $a = 0, 1$ and stationary joint-product fuel, the exact projected PCA satisfies Theorem 4.1 with its perturbation parameter equal to $p(\varepsilon)$. Consequently $\mathbb{P}(b_t \ne a) \le 96p(\varepsilon)$ at every physical site and time. Cesàro limits give two translation-invariant stationary laws with separated bit marginals, since $192\tau < 1$. Every I_j is therefore nonergodic.

For $1 \le j < n$ set

$$s_j = L^{-[4(n-j)+1]}, \qquad J_j = (s_j/2, 2s_j).$$

At $\varepsilon \in J_j$, a pair $\ell \le j$ can open only if its short track is zero. Its short capacity is at least $L^{4(n-j)+2}$, so its opening probability is less than $2/L$. A pair $\ell > j$ can open only if its long track is positive. Its long capacity is at most $L^{4(n-j)}$, so its opening probability is less than $1/L$. Thus

$$g(\varepsilon) < (n+j)/L \le 2n/L, \qquad \lambda(\varepsilon) := 3(1-\varepsilon)g(\varepsilon) < 6n/L < 1.$$

All these estimates use union bounds, not independent tracks. Common-noise coupling erases initial fuels by K; each later bit discrepancy needs an open unreplaced majority-parent path. The endpoint lemma's proof, with three parents and opening probability $g(\varepsilon)$, gives

$$\sup_x d_{\mathrm{TV}}((\delta_x P^{K+t})_W, \mu_W) \le |W|\lambda(\varepsilon)^t.$$

Hence every J_j is uniformly exponentially ergodic. This is a full-process estimate and requires no independence between heights.

The intervals lie in $(0,1)$, and $s_j/w_j = w_{j+1}/s_j = L^2 > 4$ gives their strict order

$$I_1 < J_1 < I_2 < \cdots < J_{n-1} < I_n.$$

For $n = 1$ there are no separators. Finally $g(\varepsilon) \le \sum_j \mathbb{P}(h_{2j-1} > 0) \le \varepsilon nK/2$. The same path coupling gives endpoint ergodicity for $0 < \varepsilon < 1/(3nK)$ and $2/3 < \varepsilon \le 1$. These ranges lie below I_1 and above I_n, respectively. □

Remark 7.2 (Infinite windows remain open). This theorem supplies a different finite-alphabet rule for each n. It does not give one rule with infinitely many alternating windows. There is a precise limitation of these certificates for finite tracks. Conditioning on the common reset age a, independent reset marks give

$$G(a) = 1 - \prod_{j=1}^{n} \left[1 - \frac{(H_{2j-1} - a)_+}{H_{2j-1} + 1} \frac{\min(a, H_{2j}) + 1}{H_{2j} + 1} \right].$$

As $G(a) = 0$ for $a \geq K$,

$$g(\varepsilon) = \sum_{a=0}^{K-1} \varepsilon(1-\varepsilon)^a G(a)$$

is a rational-coefficient polynomial. So are $p(\varepsilon) = 1 - (1-\varepsilon)g(\varepsilon)$ and $\lambda(\varepsilon) = 3(1-\varepsilon)g(\varepsilon)$. Their strict certificate sets $p < \tau$ and $\lambda < 1$ have finitely many components: a nonzero boundary polynomial has finitely many roots; an identically zero one gives no strict certificate on its boundary. This is a finiteness statement about these two sufficient certificates, not about the actual phase diagram. A single nilpotent finite-alphabet automaton with infinitely many alternating nonergodic windows remains open here.

8 Separated deterministic replacement events

A replacement event is a point $(u, z, s) \in \mathbb{Z}^2 \times \mathbb{Z} \times \mathbb{N}_{\geq 1}$ at which, after the deterministic update into time s, a prescribed symbol overwrites the output. All other updates follow F_H. Distances between events are measured in all four coordinates using ℓ^∞. In particular, separation in the z coordinate suffices even when events have the same time.

Theorem 8.1. *Given $d \geq 0$, choose an integer $m > d$ and let $H = m^4$. There is a set $\mathcal{E}$ of replacement events at pairwise ℓ^∞ distance at least m such that every event writes $(0, H)$, and the two initial configurations*

$$(b_0, h_0) \equiv (0, H), \qquad (b'_0, h'_0) \equiv (1, H)$$

produce $b_t(u, 0) = 0$ and $b'_t(u, 0) = 1$ for every $u \in \mathbb{Z}^2$ and $t \geq 0$ under these common events. The underlying rule is nilpotent of exponent $H + 1$.

Proof. For $u = (x, y)$ and $t \geq H$, let $[x]_m, [y]_m, [t]_m \in \{0, \ldots, m-1\}$ be their residues, and set

$$c(u, t) = [x]_m + m[y]_m + m^2[t]_m, \qquad a(u, t) = m\, c(u, t). \tag{27}$$

The age satisfies $0 \leq a(u, t) \leq m^4 - m = H - m$. Place one event for each such pair (u, t) at

$$E(u, t) = \big(u, 1 + a(u, t), t - a(u, t)\big), \tag{28}$$

and let every event write $(0, H)$. Its time is at least $m \geq 1$, and its height coordinate is positive, so these are valid events and none belongs to the bit plane $z = 0$.

We first check separation. If two ages in (28) differ, their z coordinates differ by a nonzero multiple of m, hence by at least m. If the ages agree, the residue triples in (27) agree. Distinct pairs (u, t) then differ in at least one of x, y, t by a nonzero multiple of m. Equal ages preserve that difference in the corresponding event coordinates $x, y, s = t - a$. The events are therefore distinct and at pairwise distance at least $m > d$.

We next check that the gates into $z = 0$ are always open. With initial countdowns equal to H, and all subsequent resets equal to H, monotonicity of the decrement map gives

$$h_t(u, 1) \geq (H - t)_+.$$

This is positive for $0 \leq t < H$. For $t \geq H$, the event $E(u, t)$ puts a countdown H on the backward diagonal of $(u, 1, t)$, at age $a = a(u, t) < H$. Propagating this value forward along that diagonal gives a countdown at least $H - a \geq m > 0$ at $(u, 1, t)$. Any intervening reset sets the countdown to H and can only increase this lower bound. Hence every gate into the bit plane is open at every time.

The bits at $z = 0$ are never replaced and read bits only in that plane. They consequently follow exact NEC majority forever. The two constant initial bit planes remain respectively zero and one, proving the claimed discrepancy. Nilpotency of the rule without replacements is Proposition 2.1. □

The separation parameter is fixed before the automaton is chosen. Thus the theorem provides arbitrarily well separated renewal events across the family of nilpotent rules. Their positions in the extra spatial coordinate keep the renewals separated while feeding every gate of the same bit plane.

9 Nilpotency in two steps

The rules of the preceding sections erase a configuration only after a long countdown; the rule of Theorem 4.2 needs $2^{88} + 1$ steps. The least nilpotency time that permits a negative answer to the question of [2] is two.

Proposition 9.1 (One step forces ergodicity)**.** *Let F be a cellular automaton whose image is a single configuration. Every perturbation of F by zero-range noise has, after one step and from every initial law, one and the same product law. This law is its unique stationary law.*

Proof. The image configuration is invariant under translations, since F commutes with them, so it is uniform with some symbol c. A zero-range perturbation applies F and then, independently at every site, one fixed stochastic matrix θ to the resulting symbol [2]. After one step the configuration therefore has the product law with one-site marginal $\theta(c, \cdot)$, whatever the initial law. This law is stationary, and every stationary law equals it. □

Two steps suffice. In place of a countdown, each site reads a *seed*: the word formed by the auxiliary symbols at the m sites above it. Every deterministic update resets the auxiliary symbols to zero. After one step every seed is therefore the zero word, and the second step writes the data symbol attached to that word at every site. Under noise, replacements write new auxiliary symbols at every step, and the seed read by a plane is randomness that the plane has not used before. As in Section 6, the extra coordinate supplies this randomness.

9.1 Seeded rules

Definition 9.2 (Seeded rule)**.** Let $\mathcal{B}$ be a finite alphabet and $N \subset \mathbb{Z}^d$ a finite nonempty set. Let $g_1, \ldots, g_B$ be maps from $\mathcal{B}^N$ to $\mathcal{B}$, of which g_1 is constant with value s_0. Let $S \geq 2$ and $m \geq 1$ be integers, and let $J \colon \{0, \ldots, S-1\}^m \to \{1, \ldots, B\}$ send the zero word to 1. The *seeded rule*

with these data is the cellular automaton F on $\mathbb{Z}^d \times \mathbb{Z}$ with alphabet $\mathcal{B} \times \{0, \dots, S-1\}$ given, for a configuration (b, y) and a site (u, z), by

$$(Fy)(u,z) = 0, \tag{29}$$

$$(Fb)(u,z) = g_{J(W(u,z))}\big((b(u+n,z))_{n\in N}\big), \tag{30}$$

where $W(u,z) = (y(u,z+1), \dots, y(u,z+m))$ is the *seed* of (u,z). The sets $J^{-1}(i)$ are the *bins*.

The neighbourhood of F consists of $N \times \{0\}$ and the sites $(0,j)$ with $1 \le j \le m$. Its uniform replacement perturbation Φ_ε at rate $0 < \varepsilon \le 1$ is defined as in Section 2. At every site v and time $t \ge 1$ there are independent variables $R_t(v) \sim \text{Bernoulli}(\varepsilon)$, $V_t(v)$ uniform on $\mathcal{B}$ and $Y_t(v)$ uniform on $\{0, \dots, S-1\}$. If $R_t(v) = 1$, the symbol $(V_t(v), Y_t(v))$ is written at v; otherwise F is applied there. Put

$$q_\varepsilon(0) = 1 - \varepsilon + \frac{\varepsilon}{S}, \qquad q_\varepsilon(s) = \frac{\varepsilon}{S} \quad (1 \le s < S), \qquad \pi_i(\varepsilon) = q_\varepsilon^{\otimes m}\big(J^{-1}(i)\big). \tag{31}$$

Write $\langle s \rangle$ for the constant map with value s, and let

$$\Lambda_\varepsilon = \frac{\varepsilon}{|\mathcal{B}|} \sum_{s \in \mathcal{B}} \delta_{\langle s \rangle} + (1-\varepsilon) \sum_{i=1}^{B} \pi_i(\varepsilon)\, \delta_{g_i},$$

a law on maps $\mathcal{B}^N \to \mathcal{B}$.

Lemma 9.3 (Seeded plane factor)**.** *Let F be a seeded rule and $0 < \varepsilon \le 1$. Run Φ_ε from an initial configuration, possibly random, that is independent of the noise.*

(i) *$F \circ F$ maps every configuration to the uniform configuration $(s_0, 0)$.*

(ii) *For $t \ge 1$, each auxiliary symbol $y_t(v)$ is a function of $R_t(v)$ and $Y_t(v)$ alone. These symbols are independent with law q_ε.*

(iii) *Fix a height z. For $u \in \mathbb{Z}^d$ and $t \ge 1$, let $G_t(u) = \langle V_t(u,z) \rangle$ if $R_t(u,z) = 1$, and $G_t(u) = g_{J(W_{t-1}(u,z))}$ otherwise, where W_{t-1} denotes the seed at time $t-1$. Then $b_t(u,z) = G_t(u)\big((b_{t-1}(u+n,z))_{n\in N}\big)$. The maps $G_t(u)$ with $u \in \mathbb{Z}^d$ and $t \ge 2$ are independent with law Λ_ε. For every $s \ge 1$, those with $t > s$ are independent of the data $b_0(\cdot,z), \dots, b_s(\cdot,z)$ of the plane. If the initial auxiliary symbols are independent with law q_ε and independent of the initial data, the same holds for $t \ge 1$ and $s \ge 0$.*

(iv) *Let $\pi_{\mathrm{nc}}(\varepsilon)$ be the sum of $\pi_i(\varepsilon)$ over the indices i for which g_i is not constant, and put $\lambda(\varepsilon) = |N|(1-\varepsilon)\pi_{\mathrm{nc}}(\varepsilon)$. Then*

$$\lambda(\varepsilon) \le |N| \min\big(1-\varepsilon,\ m\varepsilon(1-1/S)\big). \tag{32}$$

If $\lambda(\varepsilon) < 1$, then Φ_ε has a unique stationary law μ and, for every finite $W \subset \mathbb{Z}^{d+1}$ and $n \ge 1$,

$$\sup_x d_{\mathrm{TV}}\big((\delta_x \Phi_\varepsilon^{1+n})_W, \mu_W\big) \le |W| \lambda(\varepsilon)^n. \tag{33}$$

In particular Φ_ε is uniformly exponentially ergodic for $0 < \varepsilon < 1/[|N|m(1-1/S)]$ and for $1 - 1/|N| < \varepsilon \le 1$.

Proof. (i) By (29), every auxiliary symbol of $F(x)$ is zero. In the next update every seed is the zero word, which lies in the first bin, so (30) writes s_0 at every site. The uniform configuration $(s_0, 0)$ is fixed by F.

(ii) The deterministic update writes 0 and a replacement writes $Y_t(v)$. Thus $y_t(v) = Y_t(v)$ if $R_t(v) = 1$ and $y_t(v) = 0$ otherwise, which has law q_ε.

(iii) The formula for b_t is the data coordinate of the update. By (ii), for $t \geq 2$ the map $G_t(u)$ is a function of $R_t(u,z)$ and $V_t(u,z)$ together with $R_{t-1}(u,z+j)$ and $Y_{t-1}(u,z+j)$ for $1 \leq j \leq m$. For distinct pairs (u,t) these sets of variables are disjoint: the first two lie at height z and the others strictly above it, and they carry the indices (u,t) and $(u,t-1)$ respectively. Hence the maps are independent. If $R_t(u,z) = 1$ the map is $\langle V \rangle$ with V uniform. If $R_t(u,z) = 0$, the seed has law $q_\varepsilon^{\otimes m}$ by (ii), so the map is g_i with probability $\pi_i(\varepsilon)$. This is the law Λ_ε.

The data of the plane up to time s are functions of the initial configuration, of the variables $R_r(\cdot,z), V_r(\cdot,z)$ with $1 \leq r \leq s$, and of the variables $R_r(\cdot,z'), Y_r(\cdot,z')$ with $z' > z$ and $1 \leq r \leq s-1$, through the seeds at times $1,\dots,s-1$. The maps G_t with $t > s$ use only variables at height z with times at least $s+1$, and variables above z with times at least s. These sets of variables are disjoint, and the noise is independent of the initial configuration. Under the additional assumption on the initial auxiliary symbols, $G_1(u)$ is a function of $R_1(u,z)$, $V_1(u,z)$ and the initial seed $W_0(u,z)$, which has law $q_\varepsilon^{\otimes m}$; the initial seeds lie strictly above the plane and are independent of all initial data, and the same argument applies.

A replacement coin $R_{t-1}(u,z+j)$ also decides the data symbol at $(u,z+j)$, so different planes are correlated. The plane z never reads the data of another plane, and this correlation does not affect (iii).

(iv) Couple two initial configurations through common noise. By (ii), their auxiliary symbols agree at every time $t \geq 1$. Suppose their data differ at (u,z) at time $1+n$ with $n \geq 1$. Then the map $G_{1+n}(u)$ of the plane z is not constant, since a constant map writes the same symbol in both copies, and one of its $|N|$ inputs differs at time n. Repeating this, we obtain $w_{1+n} = u$ and $w_k \in w_{k+1} + N$ for $k = n,\dots,1$ such that the maps $G_{k+1}(w_{k+1})$, $1 \leq k \leq n$, are all nonconstant. There are at most $|N|^n$ such sequences. For a fixed sequence the n maps involved have distinct times $k+1 \geq 2$; by (iii) they are independent, and each is nonconstant with probability $(1-\varepsilon)\pi_{\mathrm{nc}}(\varepsilon)$. The union bound gives probability at most $\lambda(\varepsilon)^n$ for a data discrepancy at (u,z) at time $1+n$. This bound does not depend on the two initial configurations and does not condition on their data. Summing over W bounds the probability of a full-symbol discrepancy in W by $|W|\lambda(\varepsilon)^n$. The rest of the argument is that of Theorem 5.1: a stationary law exists by compactness, coupling with it gives (33), coupling two stationary laws gives uniqueness, and $c = -\log\lambda(\varepsilon)$ with an enlarged prefactor gives (3). If $\lambda(\varepsilon) = 0$, the two copies agree from time 2.

For (32), $\pi_{\mathrm{nc}}(\varepsilon) \leq 1$. The zero word lies in the first bin and g_1 is constant, so

$$\pi_{\mathrm{nc}}(\varepsilon) \leq 1 - \pi_1(\varepsilon) \leq 1 - q_\varepsilon(0)^m \leq m\big(1 - q_\varepsilon(0)\big) = m\varepsilon(1 - 1/S).$$

The two stated ranges make the right-hand side of (32) less than one. □

9.2 A four-symbol rule

For $m \geq 1$, let E_m be the seeded rule on $\mathbb{Z}^2 \times \mathbb{Z}$ with data alphabet $\{0,1\}$, $S = 2$, $N = \{0, e_1, e_2\}$, the maps $g_1 = \langle 1 \rangle$ and $g_2 = \mathrm{Maj}$, and J sending the zero word to 1 and every other word to 2. Its alphabet is $\{0,1\}^2$, and

$$E_m(b,y)(u,z) = \begin{cases} (1,0), & y(u,z+1) = \cdots = y(u,z+m) = 0, \\ (M(u,z),0), & \text{otherwise}, \end{cases} \tag{34}$$

where $M(u,z) = \mathrm{Maj}(b(u,z), b(u+e_1,z), b(u+e_2,z))$.

Theorem 9.4 (Two steps suffice). *Let $m = 2^{51}$. The cellular automaton E_m on $\mathbb{Z}^3$ has four symbols and neighbourhood radius 2^{51}. The map $E_m \circ E_m$ sends every configuration to the uniform configuration $(1,0)$, and E_m is not constant. For its uniform replacement perturbation at rate ε, put*

$$\rho_m(\varepsilon) = \frac{\varepsilon}{2} + (1-\varepsilon)\Big(1-\frac{\varepsilon}{2}\Big)^m.$$

(i) *For every $\varepsilon \in [2^{-44}, 2^{-42}]$ there are two translation-invariant stationary laws $\mu^0_\varepsilon, \mu^1_\varepsilon$ satisfying, at every site $v \in \mathbb{Z}^3$,*

$$\mu^0_\varepsilon\{b(v)=1\} \le 96\rho_m(\varepsilon) < 2^{-35}, \qquad \mu^1_\varepsilon\{b(v)=1\} \ge 1 - 96\rho_m(\varepsilon) > 1 - 2^{-35}.$$

(ii) *The perturbation is uniformly exponentially ergodic for $0 < \varepsilon < 1/(3\cdot 2^{50})$ and for $2/3 < \varepsilon \le 1$.*

Proof. Lemma 9.3(i) gives $E_m \circ E_m \equiv (1,0)$. The rule is not constant: with all auxiliary symbols 1 and all data 0 it writes data 0, while with all auxiliary symbols 0 it writes data 1. Its neighbourhood radius is m. Every one-site transition probability of the perturbation is at least $\varepsilon/4$, so the noise is positive and has zero range.

(i) Here $q_\varepsilon(0) = 1 - \varepsilon/2$, so the zero seed has probability $\pi_1(\varepsilon) = (1-\varepsilon/2)^m$. Fix $a \in \{0,1\}$, start from the configuration with every symbol equal to $(a,1)$, and fix a plane. At time 1 every seed is the word $(1,\dots,1)$, so each update is the majority unless it is replaced. It disagrees with the majority with probability $\varepsilon/2$, independently over the sites. For $t \ge 2$, by Lemma 9.3(iii) and conditionally on the plane's past, the updates are independent. An update disagrees with the majority with probability $\varepsilon/2 + (1-\varepsilon)\pi_1(\varepsilon) = \rho_m(\varepsilon)$ when the majority is 0, through a replacement writing 1 or a zero seed, and with probability $\varepsilon/2$ when the majority is 1, since only a replacement can write 0. Each plane is therefore a $\rho_m(\varepsilon)$-perturbation of north-east-centre majority, in the sense of Theorem 4.1, started from the constant a.

For $\varepsilon \in [2^{-44}, 2^{-42}]$ we have $\varepsilon/2 \le 2^{-43}$ and

$$\Big(1-\frac{\varepsilon}{2}\Big)^m \le \big(1-2^{-45}\big)^{2^{51}} \le e^{-64} < 2^{-64}.$$

Hence $\rho_m(\varepsilon) < 2^{-43} + 2^{-64} < 2^{-42} < \tau$, using $32\cdot 24^8 < 2^{42}$ from the proof of Theorem 4.2. Theorem 4.1 gives $\mathbb{P}\{b_t(v) \ne a\} \le 96\rho_m(\varepsilon)$ at every site and time, and $96\cdot 2^{-42} = 3\cdot 2^{-37} < 2^{-35}$. As in the proof of Theorem 4.2, Cesàro limits of the two runs are stationary and translation invariant, the one-site bounds pass to the limits, and the two laws differ because $2^{-35} < 1/2$.

(ii) Only $g_2 = \mathrm{Maj}$ is nonconstant and $|N| = 3$. Lemma 9.3(iv) with $S = 2$ gives uniform exponential ergodicity for $0 < \varepsilon < 2/(3m) = 1/(3\cdot 2^{50})$ and for $2/3 < \varepsilon \le 1$. □

Since $1/(3\cdot 2^{50}) < 2^{-44}$ and $2^{-42} < 2/3$, the rule $E_{2^{51}}$ is ergodic, non-ergodic and ergodic again as the rate increases, like F_{H_*} in Corollary 5.2. The rule F_{H_*} has radius one and is nilpotent after $2^{88}+1$ steps; $E_{2^{51}}$ is nilpotent after two steps and has radius 2^{51}. In this mechanism the radius must be large compared with $1/\varepsilon$, since the zero seed has probability $(1-\varepsilon/2)^m$ and writes 1 whatever the majority. The plane process of E_m errs toward 1 more often than toward 0, so it is not the symmetric noisy majority process. Exact bins remove this asymmetry.

9.3 The seed compiler

A *binary PCA* P on $\mathbb{Z}^d$ is given by a finite set $N \subset \mathbb{Z}^d$ and a function $p\colon \{0,1\}^N \to [0,1]$. Given $\sigma \in \{0,1\}^{\mathbb{Z}^d}$, the next configuration has independent sites, and the site u takes the value 1 with probability $p((\sigma(u+n))_{n\in N})$. Its transition probabilities are the numbers $p(x)$ and $1-p(x)$. It is *attractive* if p is nondecreasing for the coordinatewise order on $\{0,1\}^N$. Stationary laws, ergodicity and uniform exponential ergodicity are defined as in Section 2. Write $\mathbf{a}$ for the configuration with every symbol equal to a.

Lemma 9.5 (Exact seed partition)**.** *Let $S \geq 3$, $w \geq 1$, $m \geq 1$ and $B \geq 1$ be integers, put $D = w + S - 1$, and let q be the law on $\{0,\dots,S-1\}$ with $q(0) = w/D$ and $q(s) = 1/D$ for $s \neq 0$. Let $d_1,\dots,d_B$ be nonnegative integers with $\sum_i d_i = D^m$ and $d_1 \geq w^m$, and suppose*

$$m(S-1) \geq B(w-1). \tag{35}$$

Then $\{0,\dots,S-1\}^m$ has a partition into sets $\mathcal{W}_1,\dots,\mathcal{W}_B$ with $q^{\otimes m}(\mathcal{W}_i) = d_i/D^m$ for every i and with the zero word in $\mathcal{W}_1$. The partition is computable from S, w, m and $d_1,\dots,d_B$.

Proof. A word with exactly k zero letters has probability w^k/D^m, and there are $C_k = \binom{m}{k}(S-1)^{m-k}$ such words. The binomial theorem gives $\sum_k C_k w^k = D^m$. It suffices to find nonnegative integers $c_{i,k}$, for $1 \leq i \leq B$ and $0 \leq k \leq m$, with

$$\sum_i c_{i,k} = C_k, \qquad \sum_k c_{i,k} w^k = d_i, \qquad c_{i,m} = \mathbf{1}_{\{i=1\}}. \tag{36}$$

Then $\mathcal{W}_i$ takes $c_{i,k}$ of the words with k zero letters, for instance consecutive ones in lexicographic order, and the zero word, the only word with m zero letters, goes to $\mathcal{W}_1$.

The ratio $C_{k+1}/C_k = (m-k)/[(k+1)(S-1)]$ decreases in k, so on $\{0,\dots,m-1\}$ the minimum of C_k is attained at an endpoint. Since $S-1 \geq 2$, $C_0 = (S-1)^m \geq m(S-1) = C_{m-1}$. With (35),

$$C_k \geq B(w-1) \qquad (0 \leq k \leq m-1). \tag{37}$$

Put $e_i^{(0)} = d_i - w^m \mathbf{1}_{\{i=1\}}$. These are nonnegative integers with $\sum_i e_i^{(0)} = \sum_{l=0}^{m-1} C_l w^l$. Suppose that for some $0 \leq k \leq m-2$ nonnegative integers $e_i^{(k)}$ satisfy

$$\sum_i e_i^{(k)} = \sum_{l=k}^{m-1} C_l w^{l-k}. \tag{38}$$

Let $\kappa_i \in \{0,\dots,w-1\}$ be the remainder of $e_i^{(k)}$ modulo w. Then $\sum_i \kappa_i \leq B(w-1) \leq C_k$ by (37), and $\sum_i \kappa_i \equiv \sum_i e_i^{(k)} \equiv C_k$ modulo w by (38). Hence $N_k = (C_k - \sum_i \kappa_i)/w$ is a nonnegative integer. Moreover

$$\sum_i \Big\lfloor \frac{e_i^{(k)}}{w} \Big\rfloor = \frac{\sum_i e_i^{(k)} - \sum_i \kappa_i}{w} = N_k + \sum_{l=k+1}^{m-1} C_l w^{l-k-1} \geq N_k.$$

Choose integers $0 \leq t_i \leq \lfloor e_i^{(k)}/w \rfloor$ with $\sum_i t_i = N_k$, for example by filling $i = 1, 2, \dots$ in turn. Set $c_{i,k} = \kappa_i + w t_i$ and $e_i^{(k+1)} = \lfloor e_i^{(k)}/w \rfloor - t_i \geq 0$. Then $\sum_i c_{i,k} = C_k$, $e_i^{(k)} = c_{i,k} + w e_i^{(k+1)}$, and (38) holds at $k+1$. At $k = m-1$ put $c_{i,m-1} = e_i^{(m-1)}$, whose sum is C_{m-1} by (38). Finally put $c_{i,m} = \mathbf{1}_{\{i=1\}}$. Unwinding $e_i^{(k)} = c_{i,k} + w e_i^{(k+1)}$ gives $\sum_{k<m} c_{i,k} w^k = e_i^{(0)}$, which is (36). Every step is a finite computation with integers. □

The lemma only allocates the counts $c_{i,k}$; the bin of a given word is obtained from its number of zero letters and its lexicographic rank among the words with that number, so the local rule is specified without listing the S^m words.

Lemma 9.6 (Monotone envelope)**.** *Let P be an attractive binary PCA on $\mathbb{Z}^d$ and put*

$$\Delta_s = \delta_{\mathbf{1}} P^s\{\sigma(0) = 1\} - \delta_{\mathbf{0}} P^s\{\sigma(0) = 1\}.$$

Then Δ_s is nonincreasing, with a limit $\Delta_\infty \geq 0$. The PCA is ergodic if and only if $\Delta_\infty = 0$. If $\Delta_\infty > 0$, it has two distinct translation-invariant stationary laws. If it is uniformly exponentially ergodic with constants C, c, then $\Delta_s \leq 2Ce^{-cs}$.

Proof. Realize all starts with common independent uniform variables $U_t(u)$ on $[0,1]$, setting $\sigma_t(u) = 1$ exactly when $U_t(u) \leq p((\sigma_{t-1}(u+n))_{n\in N})$. Since p is nondecreasing, this coupling preserves the coordinatewise order. The law after one step from $\mathbf{1}$ is carried by configurations below $\mathbf{1}$, so $\delta_{\mathbf{1}} P^{s+1}\{\sigma(0) = 1\} \leq \delta_{\mathbf{1}} P^s\{\sigma(0) = 1\}$; similarly the corresponding sequence from $\mathbf{0}$ is nondecreasing. By the order and translation invariance, $\Delta_s \geq 0$ is the probability that the coupled runs from $\mathbf{1}$ and $\mathbf{0}$ differ at a given site.

If P is ergodic, both laws converge to its stationary law, so $\Delta_\infty = 0$. Conversely, the run from any configuration lies between the runs from $\mathbf{0}$ and $\mathbf{1}$, so two runs differ at a given site at time s with probability at most Δ_s. If $\Delta_\infty = 0$, the coupling argument in the proof of Theorem 5.1 gives a unique stationary law and convergence from every start.

If $\Delta_\infty > 0$, let $\bar{\mu}$ and $\underline{\mu}$ be weak limit points of the Cesàro averages of $\delta_{\mathbf{1}} P^s$ and $\delta_{\mathbf{0}} P^s$. They are stationary by the Feller argument of Section 2 and translation invariant because the starts are. Their probabilities of $\{\sigma(0) = 1\}$ are the limits of the two monotone sequences above, which differ by Δ_∞, so $\bar{\mu} \neq \underline{\mu}$. Finally, under (3) with $W = \{0\}$, Δ_s is at most the sum of the two one-site distances to the stationary law, hence at most $2Ce^{-cs}$. □

Theorem 9.7 (Seed compiler)**.** *Fix $d \geq 1$ and rationals ρ and ε with $0 < \varepsilon < 2\rho \leq 1$. There is a total computable map $P \mapsto F_P$, defined on attractive binary PCA on $\mathbb{Z}^d$ whose transition probabilities are rational and lie in $[\rho, 1-\rho]$, to seeded rules on $\mathbb{Z}^d \times \mathbb{Z}$, with the following properties. Write Φ_P for F_P followed by uniform replacement at rate ε.*

(i) *$F_P \circ F_P$ is constant, and F_P is not.*

(ii) *Every horizontal plane of Φ_P runs P exactly: its data form a Markov chain with the transition of P from time 1, for every initial configuration, and from time 0 if the initial auxiliary symbols are independent with law q_ε and independent of the initial data.*

(iii) *Φ_P is ergodic if and only if P is ergodic.*

(iv) *Φ_P is uniformly exponentially ergodic if and only if P is.*

(v) *Φ_P has two distinct translation-invariant stationary laws if and only if P has.*

The rate ε and the dimension $d+1$ are fixed in advance; the alphabet and the radius of F_P depend on P.

Proof. Construction. Let P be given by N and p, and put $a(x) = (p(x) - \varepsilon/2)/(1-\varepsilon)$. Since $\varepsilon/2 < \rho \leq p(x) \leq 1-\rho < 1-\varepsilon/2$, every $a(x)$ is a rational number in $(0,1)$, and a is nondecreasing. Let $a_1 < \cdots < a_L$ be its distinct values and $j(x)$ the index with $a(x) = a_{j(x)}$. Put $B = L+1$,

$$r_1 = a_1, \qquad r_i = a_i - a_{i-1}\ (2 \leq i \leq L), \qquad r_B = 1 - a_L, \qquad g_i(x) = \mathbf{1}_{\{i \leq j(x)\}}\ (1 \leq i \leq B).$$

The r_i are positive rationals with sum 1, $g_1 = \langle 1 \rangle$, $g_B = \langle 0 \rangle$, every g_i is nondecreasing, and

$$\sum_{i=1}^{B} r_i \, g_i(x) = \sum_{i \le j(x)} r_i = a(x). \tag{39}$$

Write $\varepsilon = \alpha/\beta$ with integers $1 \le \alpha < \beta$, let K be the least common denominator of the r_i, and put $T = 3K$,

$$S = \alpha T, \qquad w = (\beta - \alpha)T + 1, \qquad D = \beta T = w + S - 1.$$

Since $1/D = \varepsilon/S$, the law q of Lemma 9.5 is the law q_ε of (31). As $q_\varepsilon(0) < 1$, there is a least integer $m \ge 1$ with $m(S-1) \ge B(w-1)$ and $q_\varepsilon(0)^m \le r_1$. The integers $d_i = r_i D^m$ satisfy the hypotheses of Lemma 9.5, since K divides D, and $d_1 \ge w^m$ is the second condition on m. The lemma gives J with $q_\varepsilon^{\otimes m}(J^{-1}(i)) = r_i$ and the zero word in the first bin. Let F_P be the seeded rule with data alphabet $\{0,1\}$, neighbourhood N, maps g_i and bin map J. Every step is a finite computation with rationals, so $P \mapsto F_P$ is total computable. Any other multiple T of K with $\alpha T \ge 3$, and any m satisfying the two conditions, give a rule with all the properties below; the choices above make the map single-valued.

(i) and (ii). Lemma 9.3(i) gives $F_P \circ F_P \equiv (1, 0)$. The last bin is nonempty because $r_B > 0$. A configuration whose seed at one site lies in it receives data 0 there, since $g_B = \langle 0 \rangle$, while zero auxiliary symbols give data 1; so F_P is not constant. By construction $\pi_i(\varepsilon) = r_i$. By Lemma 9.3(iii) and (39), conditionally on the past of a plane after time 1, its updates are independent and take the value 1 with probability $\varepsilon/2 + (1-\varepsilon)a(x) = p(x)$, where x is the input. This is (ii).

From P to Φ_P. Couple all initial configurations of Φ_P through common noise; their auxiliary symbols agree from time 1. Let $\bar{b}$ and $\underline{b}$ be the data processes started at time 1 from all ones and all zeros and driven by the same maps $G_t(u)$, $t \ge 2$. Every map is a constant or some g_i, hence nondecreasing, so the data of every run lie between $\underline{b}_t$ and $\bar{b}_t$ at all times $t \ge 1$. By Lemma 9.3(iii), applied from time 1, the plane z of $\bar{b}_{1+s}$ has law $\delta_{\mathbf{1}} P^s$, and that of $\underline{b}_{1+s}$ has law $\delta_{\mathbf{0}} P^s$. Hence, for every finite $W \subset \mathbb{Z}^{d+1}$ and $s \ge 0$,

$$\sup_{x,x'} \mathbb{P}\big(X_{1+s}|_W \ne X'_{1+s}|_W\big) \le |W| \Delta_s. \tag{40}$$

If P is ergodic, $\Delta_s \to 0$ by Lemma 9.6, and the coupling argument of Theorem 5.1 shows that Φ_P is ergodic. If P is uniformly exponentially ergodic, Lemma 9.6 gives $\Delta_s \le 2Ce^{-cs}$, and the same argument gives (3) for Φ_P with the constants $\max(1, 2Ce^c)$ and c.

From Φ_P to P. For $\sigma \in \{0,1\}^{\mathbb{Z}^d}$, let λ_σ be the law with data σ in every plane and independent auxiliary symbols with law q_ε. By (ii), the data of plane 0 under $\lambda_\sigma \Phi_P^t$ have law $\delta_\sigma P^t$. Suppose that Φ_P is ergodic with stationary law μ, and let ν be the law of the data of plane 0 under μ. For a continuous function f of these data, dominated convergence gives

$$(\delta_\sigma P^t)(f) = \int (\delta_x \Phi_P^t)(f) \, \lambda_\sigma(dx) \to \mu(f) = \nu(f).$$

Thus every run of P converges to ν, which is stationary because P is Feller. If ν' is stationary for P, the same computation with σ drawn from ν' shows that $\nu' = \nu' P^t$ converges to ν, so $\nu' = \nu$. Hence P is ergodic. If Φ_P is uniformly exponentially ergodic, joint convexity of the total variation distance gives, for finite $W \subset \mathbb{Z}^d$,

$$d_{\mathrm{TV}}\big((\delta_\sigma P^t)_W, \nu_W\big) \le \int d_{\mathrm{TV}}\big((\delta_x \Phi_P^t)_{W \times \{0\}}, \mu_{W \times \{0\}}\big) \lambda_\sigma(dx) \le C|W| e^{-ct},$$

so P is uniformly exponentially ergodic. This proves (iii) and (iv).

(v). Let ν be a translation-invariant stationary law of P. Start Φ_P with independent copies of ν in the planes and with independent auxiliary symbols of law q_ε, independent of the data. This initial law is translation invariant on $\mathbb{Z}^{d+1}$. By (ii), each plane runs P from ν, so its data have law ν at every time. Every weak limit point of the Cesàro averages is stationary and translation invariant, and its data in each plane have law ν. Two distinct translation-invariant stationary laws of P therefore give two distinct ones of Φ_P. Conversely, if Φ_P has two distinct stationary laws, it is not ergodic; then P is not ergodic by (iii), and Lemma 9.6 gives two distinct translation-invariant stationary laws of P. □

Remark 9.8 (What uses attractiveness). Attractiveness is used in the passage from P to Φ_P in (iii) and (iv), and in the converse direction of (v), through Lemma 9.6. The construction, (ii), the passage from Φ_P to P in (iii) and (iv), and the lift of translation-invariant stationary laws from P to Φ_P in (v) hold for every binary PCA with rational transition probabilities in $(\varepsilon/2, 1-\varepsilon/2)$, with the maps g_i defined from the order of the values of a as above. For a non-attractive PCA, ergodicity of every plane need not give ergodicity of the whole system, since the planes read overlapping seeds; our argument does not establish the remaining directions for non-attractive PCA. Representing a PCA by independent random local maps is standard. What the compiler adds is that this randomness is read from the replacement noise of a deterministic rule at one fixed rate, with exact rational bins, and that the rule is nilpotent in two steps.

9.4 Noisy majority and the halting family

For $0 < \delta < 1$, let T_δ be noisy north-east-centre majority on $\mathbb{Z}^2$, the binary PCA with

$$p(x) = (1-\delta)\,\mathrm{Maj}(x_0, x_{e_1}, x_{e_2}) + \delta/2.$$

It is attractive, and its transition probabilities lie in $[\delta/2, 1-\delta/2]$.

Corollary 9.9 (Compiled noisy majority)**.** *Let $\delta = 2^{-42}$, $\varepsilon = 2^{-43}$ and $r = 1/(2^{44}-2)$. Compiling T_δ at rate ε as in the proof of Theorem 9.7, with $T = 2^{44}-2$ and $m = 2^{50}$, gives a cellular automaton F on $\mathbb{Z}^3$ with $2^{45}-4$ symbols and neighbourhood radius 2^{50}, such that $F \circ F$ is constant. Its data update writes* 1*, the majority of the three inputs in the plane, or* 0*, according as the seed lies in one of three bins of probabilities r, $1-2r$ and r under $q_\varepsilon^{\otimes m}$. Under uniform replacement at rate ε:*

(i) *every horizontal plane runs T_δ exactly from time* 1 *for every initial configuration, and from time* 0 *if the initial auxiliary symbols are independent with law q_ε and independent of the initial data; and there are two translation-invariant stationary laws μ^0, μ^1 with $\mu^0\{b(v) = 1\} \le 96 \cdot 2^{-43} < 2^{-36}$ and $\mu^1\{b(v) = 1\} \ge 1 - 2^{-36}$ at every site v;*

(ii) *the perturbation is non-ergodic at every rate in an open interval containing ε, and uniformly exponentially ergodic at every rate in $(0, 1/(3 \cdot 2^{50}))$ and in $(2/3, 1]$.*

Proof. Here $\varepsilon < \delta$, so Theorem 9.7 applies with $\rho = \delta/2$. The function $a = (p - \varepsilon/2)/(1-\varepsilon)$ takes the two values

$$a_1 = \frac{2^{-43} - 2^{-44}}{1 - 2^{-43}} = \frac{1}{2^{44}-2} = r, \qquad a_2 = 1 - r,$$

so $B = 3$, the bin probabilities are r, $1-2r$ and r, and $g_1 = \langle 1 \rangle$, $g_2 = \mathrm{Maj}$, $g_3 = \langle 0 \rangle$. Since $1 - 2r = (2^{43}-2)/(2^{43}-1)$, the least common denominator is $K = 2^{44}-2$, and we take

$T = K$. With $\alpha = 1$ and $\beta = 2^{43}$,

$$S = 2^{44} - 2, \qquad w = (2^{43} - 1)(2^{44} - 2) + 1, \qquad D = 2^{43}(2^{44} - 2),$$

and the alphabet $\{0,1\} \times \{0, \dots, S-1\}$ has $2S = 2^{45} - 4$ symbols. Condition (35) holds: $m(S-1) = 2^{50}(2^{44} - 3) > 2^{93}$, while $B(w-1) = 3(2^{43} - 1)(2^{44} - 2) < 3 \cdot 2^{87} < 2^{89}$. The zero word fits in the first bin: $1 - q_\varepsilon(0) = \varepsilon(1 - 1/S) \geq 2^{-44}$, so $q_\varepsilon(0)^m \leq \exp(-2^{50} \cdot 2^{-44}) = e^{-64} < 2^{-64} < r$. Thus $T = K$ and $m = 2^{50}$ are admissible in the construction. Theorem 9.7(i) gives $F \circ F \equiv (1, 0)$, and (ii) gives the exact planes; directly, $\varepsilon/2 + (1 - \varepsilon)r = 2^{-44} + 2^{-44} = 2^{-43} = \delta/2$, because $(1 - \varepsilon)r = 2^{-44}$.

(i) Start from data $\mathbf{a}$ in every plane and independent auxiliary symbols with law q_ε. Each plane runs T_δ from $\mathbf{a}$, which is a 2^{-43}-perturbation of north-east-centre majority in the sense of Theorem 4.1. As $2^{-43} < \tau$, the bound $96 \cdot 2^{-43} = 3 \cdot 2^{-38} < 2^{-36}$ holds at every site and time, and the Cesàro argument of Theorem 4.2 gives the two laws.

(ii) At a rate ε', Lemma 9.3(iii) shows that each plane, started from data $\mathbf{a}$ and independent auxiliary symbols with law $q_{\varepsilon'}$, is a binary PCA whose updates disagree with the majority with probability $\varepsilon'/2 + (1 - \varepsilon')\pi_1(\varepsilon')$ when the majority is 0 and $\varepsilon'/2 + (1 - \varepsilon')\pi_3(\varepsilon')$ when it is 1. Each $\pi_i(\varepsilon')$ is a finite sum of products of the affine functions $q_{\varepsilon'}(s)$ of ε', hence continuous. At $\varepsilon' = \varepsilon$ both error probabilities equal $2^{-43} < \tau$. They therefore stay below τ on an open interval of rates containing ε, and the argument of (i) gives two translation-invariant stationary laws there. Finally, Lemma 9.3(iv) with $|N| = 3$ gives uniform exponential ergodicity for $0 < \varepsilon' < 1/[3 \cdot 2^{50}(1 - 1/S)]$, a range containing $(0, 1/(3 \cdot 2^{50}))$, and for $2/3 < \varepsilon' \leq 1$. □

The four-symbol rule of Theorem 9.4 reaches the same conclusion with a far smaller alphabet. The compiled rule has the additional property that each of its planes is exactly the symmetric noisy majority process $T_{2^{-42}}$ from time 1.

Remark 9.10 (A two-step halting family). The compiler also accepts a prescribed rational law of local maps, provided it gives positive weight to a constant map. Let F_M be the quorum rule of Section 6, with local rule f_M on the alphabet $\mathcal{B}_M$ of k_M symbols, and put $\varepsilon_c = \varepsilon_0/2$ with $\varepsilon_0 = 490000^{-99}$. Uniform replacement at rate ε_0 after F_M applies, independently at each site, the map f_M with probability $1 - \varepsilon_0$ and a uniform constant map otherwise. Its law of maps is

$$\frac{\varepsilon_c}{k_M} \sum_{s \in \mathcal{B}_M} \delta_{\langle s \rangle} + (1 - \varepsilon_c)\Big(\frac{1 - \varepsilon_0}{1 - \varepsilon_c}\, \delta_{f_M} + \frac{\varepsilon_0 - \varepsilon_c}{(1 - \varepsilon_c)k_M} \sum_{s \in \mathcal{B}_M} \delta_{\langle s \rangle}\Big),$$

and the law in parentheses is rational with positive weights summing to one. Apply Lemma 9.5, with parameters chosen as in the proof of Theorem 9.7, with one bin for f_M and one bin for each constant $\langle s \rangle$, the zero word in the bin of $\langle \perp \rangle$, and seeds of length m_M in the ninth coordinate. This gives a seeded rule G_M on $\mathbb{Z}^9$, computable from M, with $G_M \circ G_M$ constant. By Lemma 9.3(iii), uniform replacement at the single rate ε_c drives every plane by independent maps with exactly the displayed law, that is, by F_M under uniform replacement at rate ε_0. Retain the f_M-bin label separately from the constant-bin labels, even if their maps coincide. Designate a point faulty when it is physically replaced or, without physical replacement, its seed selects one of the constant-bin labels. In each plane, after the one-step burn-in, these fault indicators are independent with probability ε_0. Hence (19) holds with $\gamma = \varepsilon_0$. The proof of Theorem 6.1 then applies with a burn-in of one step in place of H_0, and with independent auxiliary symbols of law q_{ε_c} in place of stationary fuels: the auxiliary symbols of all runs agree from time 1, and the future maps of a plane are independent of its data at time 1. If M halts, the perturbation is uniformly exponentially ergodic; if not, it has

two translation-invariant stationary laws. So ergodicity at the single rate ε_c is undecidable even for rules that are nilpotent in two steps.

This does not improve Theorem C, and the two results are incomparable. The rule G_M is nilpotent in two steps rather than $H_0 + 1$, but it is classified at one rate rather than on the interval I_0, and its radius is not one. Its seed length m_M grows with M, as its alphabet does. The zero word has probability $q_{\varepsilon_c}(0)^{m_M}$ and must lie in the bin of $\langle \perp \rangle$, whose probability $(\varepsilon_0 - \varepsilon_c)/[(1 - \varepsilon_c)k_M]$ tends to zero as k_M grows; the construction also needs $m_M(S - 1) \geq (k_M + 1)(w - 1)$, which forces m_M to be about $(k_M + 1)/\varepsilon_c$. Theorem C keeps radius one, one neighbourhood for every M, and the whole interval I_0.

10 How this work was done

AI note. The authors chose the question, directed the work and judged every result. AI models from two developers assisted. For version 1.0.0, OpenAI's GPT-6 proposed the countdown construction, and Anthropic's Claude Opus 5.5 checked it by hand. OpenAI's GPT-6.1 Sol, through Codex, searched the literature, wrote the proofs and prepared the manuscript. Anthropic's Claude Fable 5.2 read the manuscript refutation-first, in two rounds, before release, and its findings were fixed.

Version 1.1.0 adds Sections 6 and 7. OpenAI's GPT-6.1 Sol derived them, and separate agents of the same model family audited the lift, the imported interface and the window construction refutation-first. Anthropic's Claude Opus 5.5 then checked both sections refutation-first, as a review from the other developer. It found no error in the mathematics and suggested presentational changes, which this version makes; a separate run of Claude Opus 5.5 integrated the sections and those changes into the manuscript.

Version 1.2.0 adds Section 9. OpenAI's GPT-6.1 Sol derived the seed partition of Lemma 9.5, the seed compiler of Theorem 9.7 and the two-step halting family of Remark 9.10. Separate agents of the same model family audited them, and two independently written programs checked the integer allocation of Lemma 9.5 by exact arithmetic. Anthropic's Claude Opus 5.5 then checked these results refutation-first, as a review from the other developer. It re-derived the identities by hand, found no error, observed that the halting family is incomparable with Theorem C rather than an improvement of it, noted that one deterministic step cannot suffice (Proposition 9.1), and worked out the application of the compiler to noisy majority (Corollary 9.9). A separate run of Claude Opus 5.5 wrote Section 9: it derived the four-symbol rule of Theorem 9.4, the explicit constants of Corollary 9.9, the ergodic ranges of Lemma 9.3, Theorem 9.4, and Corollary 9.9, and the proofs as written, and integrated them into the manuscript. A separate run of OpenAI's GPT-6.1 Sol then checked Section 9 refutation-first; it found no error in its results, and its four corrections of statement are made here.

Models are not listed as authors; the authors take responsibility for every claim. The proofs of the countdown reduction, coupling bound, separated-event construction, countdown lift, window construction, seed partition, seed compiler and two-step rules are given in full, and none of them uses a computation. The majority-stability estimate is imported from [1]; its specialization and constants are stated in Theorem 4.1. Section 6 and Remark 9.10 also import the machine-to-tiling compiler, the three-token theorem and the Common local maps lemma of [7]; the proof of the three-token theorem there includes one finite lemma that is checked by computer [7, Lemma 9.12].

Verification status. Sections 2 to 8 were checked as described above. In Section 9, Lemma 9.5, Theorem 9.7 and Remark 9.10 were checked by models of both developers. The reviewer also ran the allocation of Lemma 9.5 by exact arithmetic, enumerating every seed word on small instances, and every bin received exactly its target mass; like the two programs above, these runs are checks and are used in no proof. Theorem 9.4, the constants of Corollary 9.9 and the ergodic ranges in Section 9 were derived while this version was prepared, and were then checked refutation-first, with independently written programs, by OpenAI's GPT-6.1 Sol as a review from the other developer. That review found no error in the results and four points of statement, which this version corrects.

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